Problem Set 1 (Analytical Part)

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September 22, 2026

Question 1

Suppose without loss of generality:

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} = \begin{bmatrix} \boldsymbol{a}_1^\top \\ \boldsymbol{a}_2^\top \\ \vdots \\ \boldsymbol{a}_n^\top \end{bmatrix}; \quad \boldsymbol{B} = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nn} \end{bmatrix} = \begin{bmatrix} \boldsymbol{b}_1 & \boldsymbol{b}_2 & \cdots & \boldsymbol{b}_n \end{bmatrix}$$

$$\forall i \in \{1, 2, \cdots, n\}, \quad \boldsymbol{a}_i^\top = \begin{bmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{bmatrix}; \quad \boldsymbol{b}_i = \begin{bmatrix} b_{1i} \\ b_{2i} \\ \vdots \\ b_{ni} \end{bmatrix}$$

Thus, we have:

$$\boldsymbol{C} = \boldsymbol{AB} = \begin{bmatrix} \boldsymbol{a}_1^\top \boldsymbol{b}_1 & \boldsymbol{a}_1^\top \boldsymbol{b}_2 & \cdots & \boldsymbol{a}_1^\top \boldsymbol{b}_n \\ \boldsymbol{a}_2^\top \boldsymbol{b}_1 & \boldsymbol{a}_2^\top \boldsymbol{b}_2 & \cdots & \boldsymbol{a}_2^\top \boldsymbol{b}_n \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{a}_n^\top \boldsymbol{b}_1 & \boldsymbol{a}_n^\top \boldsymbol{b}_2 & \cdots & \boldsymbol{a}_n^\top \boldsymbol{b}_n \end{bmatrix}$$

Essentially, as shown above, the product of two $n \times n$ matrices can be decomposed into $n \times n$ inner products of two vectors of length n. Note that for $i, j \in \{1, 2, \cdots, n\}$:

$$\boldsymbol{a}_i^\top \boldsymbol{b}_j = a_{i1} b_{1j} + a_{i2} b_{2j} + \cdots + a_{in} b_{nj} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

To compute the inner product, it takes n multiplications of two scalars ($O(1)$ for each) and $n - 1$ additions:

$$n \times O(1) + O(n - 1) = O(n)$$

Since the product of $\boldsymbol{AB}$ requires computing $n \times n$ inner products of vectors, it follows that the computational complexity of such a matrix product is:

$$(n \times n) \times O(n) = O(n^3)$$

Substituting N for n, for $N \times N$ matrices $\boldsymbol{A}$ and $\boldsymbol{B}$, the computational complexity of their matrix product is $O(N^3)$.

Question 2

(a)

$$\begin{aligned}
T_{-i} &= \frac{1}{n-1}\sum_{j\neq i} X_j \\
&= \frac{1}{n-1}\left(X_1 + \cdots + X_{i-1} + X_{i+1} + \cdots + X_n\right) \\
&= \frac{1}{n-1}\left[(X_1 + \cdots + X_n) - X_i\right] = \frac{1}{n-1}\left(\sum_{j=1}^{n} X_j - X_i\right) \\
&= \frac{1}{n-1}\left(n \times \frac{1}{n}\sum_{j=1}^{n} X_j - X_i\right) \\
&= \frac{1}{n-1}\left(n\bar{X} - X_i\right) \quad \text{(By the definition of sample means)} \\
&= \frac{n\bar{X} - X_i}{n-1}
\end{aligned}$$

(b)

$$\begin{aligned}
\bar{T}_{(-)} &= \frac{1}{n}\sum_{i=1}^{n} T_{-i} = \frac{1}{n}\sum_{i=1}^{n} \frac{n\bar{X} - X_i}{n-1} \quad \text{(From part (a))} \\
&= \frac{1}{n(n-1)}\left(\sum_{i=1}^{n} n\bar{X} - \sum_{i=1}^{n} X_i\right) = \frac{1}{n(n-1)}\left(n^2\bar{X} - n \times \frac{1}{n}\sum_{i=1}^{n} X_i\right) \\
&= \frac{1}{n(n-1)}\left(n^2\bar{X} - n\bar{X}\right) \quad \text{(By the definition of sample means)} \\
&= \frac{1}{n(n-1)} \times n(n-1)\bar{X} = \bar{X}
\end{aligned}$$

(c)

$$\begin{aligned}
\widehat{\text{Bias}}_{\text{loo}} &= (n-1)\left(\bar{T}_{(-)} - T_n\right) \\
&= (n-1)\left(\bar{X} - T_n\right) \quad \text{(From part (b))} \\
&= (n-1)\left(\bar{X} - \bar{X}\right) \quad \text{(By definition)} \\
&= 0
\end{aligned}$$

(d)

$$\begin{aligned}
T_{\text{loo}} &= T_n - \widehat{\text{Bias}}_{\text{loo}} = T_n - 0 \quad \text{(From part (c))} \\
&= T_n
\end{aligned}$$

The leave-one-out procedure does not change the sample mean because the variation that arises from the loo sample mean is averaged out when we calculate the mean of all the loo sample means, which is $\bar{T}_{(-)}$.

Question 3

(a) Let $t \in [0, \infty)$ be an arbitrary real number such that $\frac{t}{n} \in [0, 1]$:

$$\Pr(nT_n > t) = \Pr\left(T_n > \frac{t}{n}\right) = \Pr\left(\min\{X_1, \cdots, X_n\} > \frac{t}{n}\right)$$

Since $X_1, \cdots, X_n \overset{i.i.d.}{\sim} \text{Unif}[0, 1]$, it follows that:

$$\begin{aligned}\Pr(nT_n > t) &= \Pr\left(\left\{X_1 > \frac{t}{n}\right\} \cap \cdots \cap \left\{X_n > \frac{t}{n}\right\}\right) \\ &= Pr\left(X_1 > \frac{t}{n}\right) \times \cdots \times Pr\left(X_n > \frac{t}{n}\right)\end{aligned}$$

Let $F(\cdot)$ be the CDF of Unif$[0, 1]$, then we have:

$$F(x) = x \quad (x \in [0, 1])$$

This further implies:

$$\begin{aligned}\Pr(nT_n > t) &= \left(1 - \Pr\left(X_1 \leq \frac{t}{n}\right)\right) \times \cdots \times \left(1 - \Pr\left(X_n \leq \frac{t}{n}\right)\right) \\ &= \left(1 - F\left(\frac{t}{n}\right)\right) \times \cdots \times \left(1 - F\left(\frac{t}{n}\right)\right) \\ &= \left(1 - \frac{t}{n}\right) \times \cdots \left(1 - \frac{t}{n}\right) \\ &= \left(1 - \frac{t}{n}\right)^n\end{aligned}$$

Note that:

$$\lim_{n\to\infty} \left(1 - \frac{t}{n}\right)^n = \exp(-t)$$

Therefore, we have:

$$\lim_{n\to\infty} \Pr(nT_n \leq t) = \lim_{n\to\infty} 1 - \Pr(nT_n > t) = 1 - \lim_{n\to\infty} \left(1 - \frac{t}{n}\right)^n = 1 - \exp(-t)$$

Note that $1 - \exp(-t)$ is exactly the CDF for the exponential distribution with rate parameter 1. Also note that $nT_n \geq 0$. Therefore, we have shown that nT_n converges in distribution to Exp(1). QED.

(b) Since there are n observations in the original sample, for any individual bootstrap draw i, the probability of drawing T_n is:

$$\Pr(X_i^* = T_n) = \frac{1}{n}$$

The probability of T_n^* not being selected is:

$$\Pr(X_i^* \neq T_n) = 1 - \Pr(X_i^*) = 1 - \frac{1}{n}$$

Given the original sample, the bootstrap draws are conditionally independent. Therefore, the

probability that T_n is not selected in the bootstrap sample is:

$$\Pr(T_n \notin \{X_1^*, \cdots, X_n^*\}) = \Pr(X_1^* \neq T_n) \times \cdots \times \Pr(X_n^* \neq T_n) = \left(1 - \frac{1}{n}\right)^n$$

By the definition of the standard limit for Euler's constant, we have:

$$\lim_{n\to\infty} \Pr(T_n \notin \{X_1^*, \cdots, X_n^*\}) = \lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^n = e^{-1}$$

(c) Since T_n is the smallest observation in the original sample, $T_n^* = T_n$ if and ony if T_n is selected at least once in the bootstrap sample.
From part (b), we know the probability that it is never selected in the bootstrap sample. As such, we have:

$$\Pr(T_n^* = T_n) = 1 - \Pr(T_n \in \{X_1^*, \cdots, X_n^*) = 1 - \left(1 - \frac{1}{n}\right)^n$$

The limiting value is:

$$\lim_{n\to\infty} \Pr(T_n^* = T_n) = 1 - \lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^n = 1 - e^{-1}$$

Note that all observations are drawn from the interval $[0, 1]$. Therefore, the true population minimum is:

$$\theta = 0$$

We have:

$$\lim_{n\to\infty} \Pr(T_n = \theta) \lim_{n\to\infty} \Pr(nT_n = 0)$$

However, from part (a), we know that nT_n converges in distribution to $\text{Exp}(1)$, which is a continuous distribution. Since the probability of a single point for a continuous distribution is zero, this implies that asymptotically:

$$\lim_{n\to\infty} \Pr(T_n = \theta) = 0$$

However, the bootstrap parallel is essentially different than the asymptotic regime as shown above, because:

$$\lim_{n\to\infty} \Pr(T_n^* = T_n) = 1 - \lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^n = 1 - e^{-1} > 0$$

In other words, asymptotically, the bootstrap distribution of T_n^* is essentially different from the sampling distribution of T_n. Therefore, the nonparametric bootstrap in this setting is not valid.

Question 4

(a) Let $Z_i = 1$ if cup i has tea poured in first and $Z_i = 0$ if milk is poured in it first. Define the potential outcome:

$$Y_i(z) = \begin{cases} 1 \text{ if would classify cup } i \text{ as tea first under } Z_i = z, \\ 0 \text{ if she would classify it as milk first.} \end{cases}$$

The sharp null hypothesis states that the true pouring order has no effect on Muriel's classification:

$$H_0 : Y_i(1) = Y_i(0) \qquad \forall i = 1, \cdots, 8$$

(b) The number of possible assignments is:

$$\binom{8}{4} = 70$$

(c) Under H_0, Muriel has selected some fixed set of four cups that she believes have received tea first. Let k be the number of the four cups in which tea is poured first, where $k \in \{0, 1, 2, 3, 4\}$.
By assumption, Muriel gets those k cups correct.
Since there are exactly four cups in which tea is poured first. The number of remaining tea-first cups is $4 - k$, which Muriel misclassified as milk-first. Those $4 - k$ tea-first cups are among the four cups that Muriel classified as milk-first. This implies that among those cups classified as milk-first, the number of actual milk-first cups that she correctly identified as milk-first is:

$$4 - (4 - k) = k$$

Therefore, the total number of cups that Muriel classified correctly is $S = k + k = 2k \in \{0, 2, 4, 6, 8\}$.
For each k, the number of possible assignments is:

$$N(k) = \binom{4}{k}\binom{4}{4-k} = \binom{4}{k}^2$$

Table 1: Possible Assignments with Different Values for S

k	Correct $S = 2k$	Number of assignments
0	0	$\binom{4}{0}^2 = 1$
1	2	$\binom{4}{1}^2 = 16$
2	4	$\binom{4}{2}^2 = 36$
3	6	$\binom{4}{3}^2 = 16$
4	8	$\binom{4}{4}^2 = 1$

(d) According to Table 1, the probability of $S = 8$ is:

$$\Pr(S = 8|H_0) = \frac{1}{70}$$

Since $S_{\text{obs}} = 8$ and 8 is the largest possible value for S, we have:

$$p = \Pr(S \geq S_{\text{obs}}|H_0) = \Pr(S \geq 8|H_0) = \Pr(S = 8|H_0) = \frac{1}{70} \approx 0.01429$$

(e) From part (d), we know:

$$p \approx 0.01429 < 0.05$$

I would reject the null hypothesis. In other words, at the 5% significance level, there is statistically significant evidence indicating that Muriel's classifications depend on the ordering of pouring tea and milk.